InvestBrand Financial Literacy Features

Product Requirements Document

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Executive Summary

This document outlines requirements for enhancing InvestBrand's brand puzzle game with integrated financial literacy features. The goal is to leverage the existing game mechanics (market index selection, sector selection, parent company selection, brand voting) to teach fundamental stock market concepts through contextual micro-lessons, progressive missions, and outcome-based learning.

The enhancement transforms InvestBrand from a pure puzzle game into an educational platform that empowers players to understand real investing principles while maintaining engagement and fun.

Background & Context

Current State

InvestBrand is a brand puzzle game where players:

1. Select a Market Index (Nifty 50, Nifty Midcap, Nifty Next 50)
2. Select a Target Sector (Automobile, Consumer, Energy, FMCG, Financials, Healthcare, IT, Industrial, Infrastructure, Materials, Metals, Telecom)

3. Select a Parent Company (e.g., Bajaj Auto, Eicher Motors, Hero MotoCorp, Mahindra & Mahindra, Maruti Suzuki, Tata Motors)
4. Vote for Core Brands (e.g., Chetak, Dominar, Pulsar for Bajaj Auto)

Each brand displays its ticker symbol (e.g., BAJAJ-AUTO).

Target Audience

Players interested in brands and puzzles who may have:

- Limited or no investing experience
- Curiosity about stock markets but find traditional resources intimidating
- Interest in learning through play rather than formal education
- Desire to connect familiar consumer brands to investment opportunities

Learning Philosophy

Research shows gamified financial education works best when it:

- Delivers one concept at a time in context
- Uses micro-lessons (1-3 sentences) rather than long explanations
- Connects game actions to real investing principles
- Provides optional depth for curious learners
- Rewards learning progression with missions and achievements

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Objectives

Primary Goals

1. Teach fundamental stock market concepts through gameplay without disrupting core game experience
2. Build financial literacy progressively from basic to intermediate concepts
3. Create sustainable engagement through educational missions and achievements
4. Bridge the gap between brand recognition and investment understanding

Success Metrics

- Player engagement with educational tooltips and cards (click-through rate > 40%)
- Mission completion rate (target > 60% for basic missions)
- Self-reported understanding of key concepts (post-game surveys)
- Retention improvement compared to baseline (pre-enhancement)
- "Learn more" content consumption rate

Feature Requirements

1. Contextual Micro-Lessons

1.1 Step-Level Tooltips

Requirement: Add educational tooltips at each of the 4 voting steps that explain the concept in 1-2 sentences.

Implementation Details:

Step	Tooltip Content
Market Index	"An index is a basket of top companies. Buying an index fund means buying the whole basket, spreading your risk across many stocks."
Target Sector	"Sectors are industry groups (Auto, IT, FMCG, etc.). Different sectors perform differently based on economic conditions. Mixing sectors reduces risk through diversification."
Parent Company	"This is the actual listed company you can invest in. Many brands belong to one parent company with a single stock ticker."
Core Brands	"These brands are owned by the parent company. Remember: you buy the company's stock (ticker), not the individual brand."

Table 1: Step-level educational tooltips

UI Behavior:

- Display tooltip as an info icon (i) next to step heading
- Tooltip appears on hover (desktop) or tap (mobile)
- Can be dismissed but reappears on next session until user marks "Got it"
- First-time users see auto-expanded tooltips for Step 1

1.2 Selection-Specific Context Chips

Requirement: When user makes a selection, show 2-3 contextual chips with relevant financial information.

Examples:

For **Nifty 50** selection:

- "Large-cap: Top 50 companies by market value"
- "Generally more stable, lower volatility"
- "Good starting point for beginners"

For **Automobile** sector selection:

- "Cyclical sector: performance tied to economic growth"
- "Affected by fuel prices, interest rates, consumer sentiment"
- "Part of Industrials in some classifications"

For **Bajaj Auto** company selection:

- "Market cap: ₹X.XX lakh crores (as of [date])"
- "Part of Nifty 50 index"
- "Ticker: BAJAJ-AUTO on NSE/BSE"

UI Behavior:

- Chips appear below selected item
- Use color coding: blue for classification, green for positive attributes, orange for risk factors
- Chips are clickable to expand full explanation
- Maximum 3 chips per selection to avoid clutter

1.3 Post-Vote "Did You Know?" Messages

Requirement: After user confirms vote, display a rotating financial literacy tip instead of generic confirmation.

Content Categories:

- Risk and diversification ("Owning one auto stock is risky; owning an index fund spreads that risk across 50 companies")
- Long-term investing mindset ("Stocks can be volatile short-term but historically grow over 10+ years")
- Fundamental concepts ("Market cap = share price × total shares; it shows company size")
- Sector knowledge ("FMCG stocks are defensive; IT stocks are growth-oriented")
- Ticker vs brand education ("You buy BAJAJ-AUTO ticker, which owns Pulsar, Dominar, Chetak brands")

UI Behavior:

- 3-5 second display window with smooth fade-in
- Dismissible with "Next" button or auto-advance
- Track which tips user has seen; prioritize unseen content
- Include "Learn more" link for deeper dive

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2. Leaderboard-Driven Learning

2.1 Performance vs Popularity Display

Requirement: After each day's puzzle, show comparison between most-voted brands and actual market performance.

Implementation:

- Daily summary card showing: "Top 3 voted brands" vs "Top 3 performers today (by % change)"
- Highlight when they align (positive reinforcement) and when they diverge (learning moment)
- Add commentary: "Popularity doesn't equal returns. Research fundamentals, not just brand love!"

Data Requirements:

- Daily stock price data for all listed parent companies
- Vote aggregation by brand/company
- Percentage change calculation (day-over-day)

UI Placement:

- Leaderboard page: new section "Yesterday's Votes vs Market"
- Optional push notification for engaged users

2.2 Strategy Profiling

Requirement: Assign players a "strategy tag" based on voting behavior, with educational context.

Strategy Tags:

Tag	Criteria	Educational Note
Index Lover	> 60% votes in Nifty 50	"You prefer stability! Index investing is beginner-friendly and low-risk."
Sector Diversifier	Votes spread across 4+ sectors in a week	"Great diversification! Mixing sectors reduces portfolio risk."
Brand Loyalist	> 70% votes in same sector	"You love a sector! Be aware of concentration risk—mix it up."
Growth Hunter	> 50% votes in IT/Healthcare	"Growth stocks can deliver high returns but are more volatile."
Value Seeker	> 50% votes in FMCG/Utilities	"Defensive stocks are stable but may grow slower than tech/IT."

Table 2: Player strategy tags and educational messages

UI Behavior:

- Display tag on player profile and leaderboard
- Update weekly based on rolling 7-day voting pattern
- Show badge with tooltip explaining strategy pros/cons
- Allow user to click for full strategy explainer page

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3. Progressive Mission System

3.1 Mission Structure

Requirement: Implement achievement-based missions that teach specific investing concepts through gameplay.

Mission Tiers:

1. **Beginner Missions** (Week 1-2): Basic concepts
2. **Intermediate Missions** (Week 3-4): Diversification, risk
3. **Advanced Missions** (Ongoing): Market dynamics, research

3.2 Beginner Missions

Mission	Completion Criteria	Reward Content
Index Explorer	Vote for at least one Nifty 50 stock from 3 different sectors	Unlock card: "Why index funds are a good starting point for beginners" with links to NSE index fund resources
Ticker Detective	Correctly match 5 brands to their listed parent company ticker	Show card on how to look up ticker symbols, market cap, sector classification on NSE/BSE websites
First Steps	Complete 5 votes (any combination)	Introduction to investment basics: "Stocks represent ownership in companies. When you buy Bajaj Auto stock, you own a tiny piece of the company."
Market Navigator	Vote in all 3 indices (Nifty 50, Midcap, Next 50) at least once	Explain large-cap vs mid-cap vs small-cap classifications and risk-return profiles

Table 3: Beginner mission specifications

3.3 Intermediate Missions

Mission	Completion Criteria	Reward Content
Diversifier	Vote for brands from 5+ different sectors in one week	Card on portfolio diversification: "Don't put all eggs in one basket. Spread investments across sectors to reduce risk."
Sector Scholar	Vote for all 12 sectors at least once	Detailed sector comparison chart with cyclical vs defensive characteristics
Consistency Champion	Vote daily for 7 consecutive days	Explain rupee-cost averaging and benefits of regular investing
Risk Awareness	Vote for mix of large-cap (Nifty 50) and mid-cap stocks in same week	Educational content on risk-return tradeoff and volatility

Table 4: Intermediate mission specifications

3.4 Advanced Missions

Mission	Completion Criteria	Reward Content
Trend Spotter	Vote for a brand whose stock gains > 3% the next day (3 times)	Introduce basic technical vs fundamental analysis concepts
Contrarian Thinker	Vote for least-popular brand that outperforms popular choices (3 times)	Content on avoiding herd mentality and doing independent research
Marathon Investor	Maintain 30-day voting streak	Long-term investing philosophy and compounding returns explanation

Table 5: Advanced mission specifications

3.5 Mission UI Components

Requirements:

- Missions tab in navigation showing all available, in-progress, and completed missions
- Progress bars showing completion percentage
- Badge/trophy visual rewards for completed missions
- Push notification when mission completed (if user opted in)
- Social sharing option: "I just earned the Diversifier badge on InvestBrand!"

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4. Optional "Learn More" System

4.1 Expandable Explainer Links

Requirement: Add "Why this matters" or "Learn more" links at each step that open detailed educational content.

Content Structure:

Each explainer should contain:

- **Concept headline** (1 sentence)

- **Why it matters** (2-3 bullets)
- **Real-world example** (1 paragraph)
- **Action takeaway** (what beginner should know/do)
- **External resources** (links to SEBI investor education, NSE Academy, Zerodha Varsity)

Example for "Market Index" explainer:

- **Headline:** "What is a market index and why should I care?"
- **Why it matters:**
  - Indices track market performance (Nifty 50 = top 50 companies)
  - Index funds let you invest in the whole basket at once
  - Lower risk than picking individual stocks
- **Example:** "The Nifty 50 includes companies like Reliance, TCS, HDFC Bank, Infosys. Instead of researching and buying 50 stocks individually, you can buy one Nifty 50 index fund and own a piece of all 50."
- **Takeaway:** "For beginners, index funds are often recommended as a starting point because they're diversified and low-cost."
- **Resources:** Links to NSE India's index methodology page, SEBI investor guide

4.2 Personalized Weekly Insights

Requirement: Send weekly email or in-app message with personalized learning content based on user's voting behavior.

Content Template:

- Greeting with player's strategy tag
- Summary of week's voting pattern (e.g., "You voted 5 times, focusing on IT and Auto sectors")
- 3 beginner investing tips tailored to their pattern
- Featured mission suggestion
- Link to relevant educational article

Example Email:

> **Subject:** Your InvestBrand Week in Review, Sector Diversifier!
> Hi Sankar,
> This week you voted 7 times across 6 different sectors—great diversification! Here are 3 investing ideas based on your play style:
> 1. **Diversification reduces risk**: You naturally mixed Auto, IT, FMCG. In real investing, spreading money across sectors protects you when one sector underperforms.
> 2. **Index funds do this automatically**: Nifty 50 includes companies from all major sectors, giving instant diversification.
> 3. **Rebalancing matters**: Over time, review if one sector dominates your portfolio and adjust.
>
> **Mission for you:** Complete "Sector Scholar" by voting in all 12 sectors!
> **Recommended read:** "Why Diversification is the Only Free Lunch in Investing" [link]

Frequency: Weekly, sent Monday mornings

Opt-out: Users can disable in settings

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5. In-Game Educational Content Database

5.1 Content Categories & Topics

Requirement: Build comprehensive content library covering:

Category	Topics
Basics	What is a stock, What is a ticker, Market cap, Share price, Dividends, Stock exchanges (NSE/BSE)
Indices	What is Nifty 50, Nifty Midcap, Nifty Next 50, Sensex, Sectoral indices
Sectors	All 12 sectors with characteristics, cyclical vs defensive, sector rotation
Risk & Return	Volatility, Risk-return tradeoff, Diversification, Concentration risk, Time horizon
Investment Vehicles	Stocks vs mutual funds vs ETFs vs index funds, Direct vs regular plans
Strategies	Long-term investing, Rupee-cost averaging, Lump-sum vs SIP, Fundamental vs technical analysis
Behavioral Finance	Herd mentality, FOMO, Loss aversion, Confirmation bias, Patience and discipline

Table 6: Educational content taxonomy

5.2 Content Format Standards

All educational content must follow:

- **Reading level:** Grade 8-10 (simple language, no jargon or explained jargon)
- **Length:** Micro-lessons 50-100 words, explainers 200-400 words, deep-dives 500-800 words
- **Structure:** Headline, bullet points, example, takeaway, links
- **Tone:** Conversational, encouraging, non-judgmental, India-specific examples
- **Visuals:** Each topic has at least one supporting chart, icon, or illustration
- **Citations:** Link to authoritative sources (SEBI, NSE, BSE, reputable financial educators)

Technical Requirements

6.1 Data Integration

- Real-time or end-of-day stock price data for all listed parent companies
- Historical price data for performance tracking
- Index composition data (which stocks are in Nifty 50, Midcap, Next 50)
- Sector classification data mapped to companies
- Vote aggregation and analytics database

Suggested APIs/Data Sources:

- NSE India official data feeds
- BSE API for stock prices
- Google Finance or Yahoo Finance for historical data
- Custom scraping with proper rate limiting and caching

6.2 User Progress Tracking

- Mission completion status per user
- Educational content viewed/clicked per user
- Tooltip dismissal tracking ("Got it" acknowledgments)
- Strategy tag calculation based on rolling 7-day window
- Weekly summary generation for personalized emails

Database Schema Additions:

- `user_missions` table: user ID, mission ID, progress, completion date
- `user_content_views` table: user ID, content ID, view timestamp
- `user_strategy_tags` table: user ID, tag, calculation date, voting pattern summary

6.3 UI/UX Components

New Components Required:

- Tooltip component with info icon, hover/tap behavior, dismissible

- Context chip component with color coding, expandable
- Mission card component with progress bar, badge, description
- "Did You Know?" modal with timer, dismissible, "Learn more" link
- Explainer page template with consistent structure
- Weekly insights email template
- Leaderboard enhancement for performance vs popularity section
- Profile enhancement for strategy tag display

Design Principles:

- Non-intrusive: educational elements should not block core game flow
- Progressive disclosure: show simple, allow opt-in to complex
- Mobile-first: all tooltips, chips, modals work perfectly on mobile
- Accessibility: proper contrast ratios, keyboard navigation, screen reader support
- Performance: educational content loads asynchronously, doesn't slow core game

6.4 Analytics & Measurement

Tracking Requirements:

- Tooltip engagement: views, dismissals, "Got it" clicks per step
- Context chip clicks and expansion rate
- "Did You Know?" message view duration and "Learn more" clicks
- Mission start, progress, completion rates by mission type
- Explainer page views, time on page, external link clicks
- Weekly email open rate, click-through rate
- Performance vs popularity section engagement
- Strategy tag assignment distribution

Reporting Dashboard:

Build admin dashboard showing:

- Overall engagement metrics (aggregated)
- Most/least engaged educational features

- Mission completion funnels
- Content effectiveness (correlation between content views and continued engagement)
- A/B test results (if running experiments on content variations)

Content Development Plan

Phase 1: Core Content (Weeks 1-2)

- Write all step-level tooltips (4 tooltips)
- Create context chips for all indices, sectors, top 20 parent companies
- Develop 20 "Did You Know?" rotating messages
- Write all beginner mission reward content (4 missions)
- Create 4 core explainer pages (Index, Sector, Diversification, Ticker/Company)

Phase 2: Extended Content (Weeks 3-4)

- Complete intermediate mission content (4 missions)
- Write sector-specific deep-dive explainers (12 sectors)
- Develop advanced mission content (3 missions)
- Create weekly email templates with variations
- Build strategy tag explainer pages (5 tags)

Phase 3: Enrichment (Weeks 5-6)

- Add 30 more "Did You Know?" messages (total 50)
- Create glossary of investing terms
- Develop video content or animations for key concepts
- Build external resource library with curated links
- Translate content to Hindi and regional languages (if applicable)

Content Review Process

1. Financial accuracy review by qualified advisor or CFA
2. SEBI compliance review (ensure no investment advice, only education)
3. Plain language review (readability testing)
4. User testing with 5-10 beginner investors

5. Iterate based on feedback

Implementation Roadmap

Sprint 1 (Weeks 1-2): Foundation

- Set up content database and CMS
- Implement tooltip component and step-level tooltips
- Add context chip component
- Integrate basic stock price data feed
- Build mission tracking database tables

Sprint 2 (Weeks 3-4): Core Features

- Implement "Did You Know?" post-vote messages
- Launch 4 beginner missions
- Create mission UI (missions tab, progress tracking)
- Add "Learn more" expandable links with 4 core explainer pages
- Build performance vs popularity leaderboard section

Sprint 3 (Weeks 5-6): Personalization

- Implement strategy tag calculation logic
- Add strategy tags to profiles and leaderboard
- Build weekly insights email system
- Launch intermediate missions (4 missions)
- Create admin analytics dashboard

Sprint 4 (Weeks 7-8): Enrichment & Polish

- Add advanced missions (3 missions)
- Complete all sector explainer pages
- Implement content recommendation engine for personalized learning
- A/B test different tooltip, chip, message variations
- User testing and iteration

Sprint 5 (Week 9+): Launch & Iterate

- Soft launch to 10% of user base
- Monitor analytics and gather feedback
- Iterate on content and UX based on data
- Full launch with marketing campaign highlighting educational value
- Ongoing content updates based on market events and user interests

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Legal & Compliance Considerations

SEBI Investor Education Guidelines

- All content must be clearly marked as "educational" and "not investment advice"
- No specific stock recommendations or "buy/sell" suggestions
- Disclaimers on every page with financial content
- Risk warnings for higher-risk concepts (individual stocks, mid-caps, etc.)

Disclaimer Language

Required on all educational pages:

> "This content is for educational purposes only and does not constitute investment advice. InvestBrand is a gaming platform. Please consult a SEBI-registered financial advisor before making investment decisions. Past performance is not indicative of future results. Investing in stocks involves risk, including potential loss of principal."

Content Accuracy

- All market data must be timestamped with "as of [date]"
- Stock prices, market cap figures updated daily or marked as indicative
- External links reviewed quarterly for accuracy and relevance
- Content review every 6 months to update for regulatory changes

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Success Criteria & Evaluation

Quantitative Metrics (6 months post-launch)

- Tooltip engagement rate > 40% (users who click/view tooltips)
- Mission completion rate > 60% for beginner missions
- "Learn more" click-through rate > 15%
- Weekly email open rate > 25%, CTR > 10%
- User retention improvement > 20% compared to pre-enhancement baseline
- Daily active users (DAU) growth > 15%

Qualitative Metrics

- User surveys showing increased understanding of investing basics
- Positive sentiment in user reviews mentioning educational value
- Media coverage highlighting InvestBrand as educational platform
- Partnerships with financial literacy organizations (SEBI, NSE Academy)

Failure Criteria (trigger for redesign)

- Tooltip engagement < 20% (content not resonating)
- Mission completion < 30% (too difficult or not compelling)
- User complaints about "too much education disrupting fun"
- Decrease in retention or DAU after launch (feature degraded experience)

Open Questions & Future Considerations

Questions for Stakeholder Review

1. Should we gate certain educational content behind achievements, or make all freely accessible?
2. What is our tolerance for adding UI elements without cluttering the interface?
3. Should we include actual investment links (e.g., "Open a demat account" CTAs) or remain purely educational?

4. Is there a partnership opportunity with brokerage platforms for user monetization?
5. Should we certify users who complete all missions (e.g., "InvestBrand Financial Literacy Certificate")?

Future Enhancement Ideas

- Portfolio simulation mode: Let users "invest" virtual currency and track performance
- Peer learning: Discussion forums or comment sections on explainer pages
- Live market events integration: Explain breaking news (stock splits, earnings, mergers) in real-time
- Gamified quizzes: Test knowledge with multiple-choice questions for bonus points
- Regional language support: Translate content to Hindi, Tamil, Telugu, Bengali for wider reach
- Partnership with financial influencers: Guest content or endorsed missions
- API for schools/colleges: Let educators use InvestBrand missions as classroom assignments

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Appendix A: Sample Content Examples

A.1 Sample Tooltip (Market Index)

> **What is a Market Index?**
> An index is a basket of top companies grouped together. For example, Nifty 50 includes the 50 largest Indian companies. Buying an index fund means buying the whole basket, which spreads your risk across many stocks instead of betting on just one.

A.2 Sample Context Chip (Nifty 50)

- Chip 1: "Large-cap: Top 50 companies by market value"
- Chip 2: "Lower volatility, good for beginners"
- Chip 3: "Diversified across all major sectors"

A.3 Sample "Did You Know?" Message

> **Did you know?** The stock market can be volatile day-to-day, but historically the Nifty 50 has grown an average of 12-15% per year over 20+ years. Patience pays in investing!

A.4 Sample Mission: Index Explorer

Title: Index Explorer
Description: Vote for at least one Nifty 50 stock from 3 different sectors to earn this badge.
Progress: 2/3 sectors completed (Auto ✓, IT ✓, FMCG pending)
Reward: Unlock the guide "Why Index Funds are Beginner-Friendly" and earn 50 bonus points.

A.5 Sample Strategy Tag: Sector Diversifier

Tag: Sector Diversifier
You earned this because: You voted across 6 different sectors this week!
What this means in real investing: Diversification reduces risk. If one sector (like Auto) underperforms due to economic slowdown, other sectors (like FMCG or IT) may hold steady or grow, protecting your overall portfolio.
Learn more: [Link to Diversification Explainer]

Appendix B: Technical Specifications

B.1 Tooltip Component Spec

- **Trigger:** Info icon (i) next to step heading
- **Behavior (Desktop):** Hover to display, click to pin, click outside to dismiss
- **Behavior (Mobile):** Tap to toggle, tap outside to dismiss
- **Position:** Above or below trigger based on viewport space
- **Max width:** 300px
- **Styling:** White background, subtle shadow, 12px padding, 14px font size
- **Dismissal:** "Got it" button sets cookie/localStorage flag to not auto-show again

B.2 Mission Progress Calculation

Example: Index Explorer Mission

```
SELECT user_id, COUNT(DISTINCT sector) as sectors_voted
FROM votes
WHERE index_name = 'Nifty 50'
AND vote_date >= CURRENT_DATE - INTERVAL '7 days'
GROUP BY user_id
HAVING COUNT(DISTINCT sector) >= 3;
```

Mission marked complete when query returns user. Trigger notification and unlock content.

B.3 Strategy Tag Algorithm

Pseudocode for "Sector Diversifier":

```
votes_last_7_days = get_user_votes(user_id, days=7)
unique_sectors = count_unique(votes_last_7_days, field='sector')

if unique_sectors >= 5:
assign_tag(user_id, tag='Sector Diversifier')
```

Recalculate weekly (Monday 00:00 IST) for all active users.

Appendix C: Content Sources & References

Recommended External Resources for Linking

- SEBI Investor Education Portal: https://investor.sebi.gov.in
- NSE Academy: https://www.nseindia.com/education
- BSE Investor Protection Fund: https://www.bseindia.com/static/investors/index.html
- Zerodha Varsity: https://zerodha.com/varsity (comprehensive free investing course)
- ET Money Learn: Articles on mutual funds, stocks, financial planning
- Moneycontrol Learn: Market basics and sector analysis

Content Review Checklist

Before publishing any educational content, verify:

- ☐ Factual accuracy reviewed by finance professional
- ☐ No investment advice or specific stock recommendations
- ☐ Disclaimer included and visible
- ☐ Reading level appropriate (Grade 8-10)
- ☐ Examples use Indian companies, rupee currency
- ☐ Links to external resources tested and valid
- ☐ Mobile responsive and accessible
- ☐ Compliant with SEBI guidelines

Document Revision History

Version	Date	Author	Changes
1.0	March 21, 2026	Sankar Ganesh Bala Krishnan	Initial document creation with all core requirements

Table 7: Document revision history

Approval Sign-Off

Role	Name	Signature & Date
Product Owner		
Engineering Lead		
Content Lead		
Legal/Compliance		

Table 8: Stakeholder approval